

GeoSenEsm

Application User Guide



*Geolocation and Sensor data supported
Experience Sampling Method*

Poznań, 17th July 2026

This document describes the features and capabilities of the GeoSenEsm application. GeoSenEsm enables the creation of periodic survey forms for respondents in studies that use the Experience Sampling Method (ESM). Respondents' answers are additionally correlated with their location and with sensor readings that measure the temperature and humidity of their immediate surroundings. The project ships a mobile application for Android and iOS. To operate a study, researchers also need to host an administrator application for creating surveys and monitoring results.

GeoSenEsm — an abbreviation of *Geolocation and Sensor data supported Experience Sampling Method* — integrates geolocation and sensor data with ESM, a widely used technique that captures participants' real-time self-reports on experiences, behaviours and emotions. By pairing subjective responses with objective environmental metrics, GeoSenEsm helps urban researchers uncover more nuanced relationships between urban environments and human behaviour.

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1 GeoSenEsm project

GeoSenEsm is open-source software distributed under the MIT licence. The full source code lives on GitHub at <https://github.com/geosenesm>.

1.1 Project highlights

Simple, inclusive design. The main screen is dedicated to surveys only. One question is shown per page and font sizes follow the device's accessibility settings.

Data ownership and privacy. All data is stored on a server operated by the researcher. Accounts cannot be self-registered, so the researcher retains full control over who joins the study.

Scheduling and notifications. Configure start/finish times per survey and repeat them multiple times a day. Respondents get a notification when a slot opens and another 15 minutes before it closes.

Offline first. Surveys, answers, GPS points and sensor readings are stored locally when the network is unavailable and pushed to the server automatically once connectivity returns.

Admin portal. Create surveys, manage sending policies, inspect results, and export CSV, JSON or ZIP from any modern browser.

Location and sensor data. Enrich responses with GPS and with temperature / humidity from Xiaomi Mi LYWSD03MMC or Kestrel Drop D2 sensors. Optionally constrain the study to a specific polygon.

Question types and conditional routes. Seven question types, section-level visibility rules (always, by group, conditional) and initial-survey grouping.

Respondent transparency. Participants can inspect their own submitted locations, live sensor readings and progress calendar inside the mobile app.

1.2 System architecture

GeoSenEsm is a multi-platform system based on the Flutter framework for the mobile client and on a Spring Boot / Angular pair on the server. Researchers provision their own server instance; the mobile app then connects to it using a URL supplied by the study administrator.

The server instance consists of a Java API, an admin web application, a relational database (SQL Server, storing the operational data) and a document store (MongoDB, mirroring full response documents for the download tab).

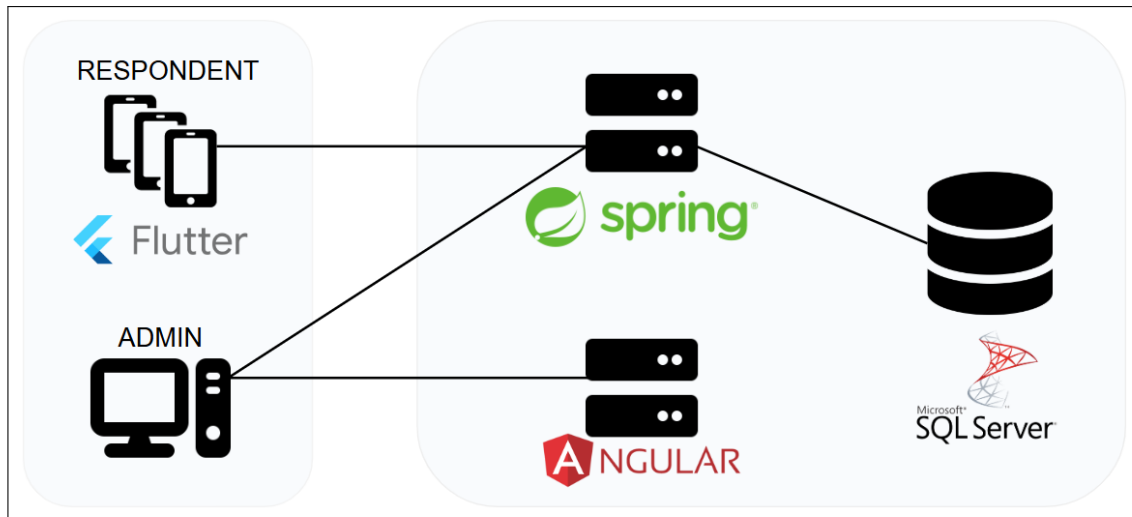


Figure 1.1: *GeoSenEsm system architecture.*

1.3 Server deployment

To set up a server, researchers need Docker installed (<https://www.docker.com/>). Docker packages applications together with their dependencies into containers; a single `docker-compose.yml` file is enough to bring up every component of GeoSenEsm without requiring specialist system-administration expertise.

Components deployed as Docker containers are:

1. the Java (Spring Boot) API,
2. the Angular admin panel,
3. SQL Server (operational database),
4. MongoDB (response-document store).

The full deployment definition and step-by-step instructions live in the DevOps repository at <https://github.com/GeoSenEsm/devops>. The only component that is not covered by `docker-compose` is the DNS / reverse-proxy configuration that maps a public domain to the API and admin containers; consult your network administrator if you need help with that step.

2 Supported sensors

2.1 Xiaomi Mi LYWSD03MMC



Figure 2.1: *Xiaomi Mi LYWSD03MMC temperature and humidity sensor.*

The Xiaomi Mi LYWSD03MMC is a widely used temperature and humidity sensor that appears on Bluetooth Low Energy scans under the fixed name LYWSD03MMC. Because the name is identical across all units, the mobile app can accidentally read from a neighbour's device. To prevent this, the admin panel exposes a MAC-address whitelist (see Section 4.11); the mobile app then connects only to the MAC assigned to the current respondent.

2.2 Kestrel Drop D2



Figure 2.2: *Kestrel Drop D2 sensor.*

The Kestrel Drop D2 is a more sophisticated device with a unique serial number that removes the ambiguity present with the Xiaomi sensor. The MAC whitelist is therefore not required for Kestrel devices.

3 Mobile application

3.1 Login and first-run setup

The first screen the user sees is the login form (Figure 3.1). Respondents fill in the server URL (*API URL*, provided by the study administrator) together with their *username* and *password*. Accounts are created centrally in the admin panel and distributed to respondents out of band (email, print-out, and so on); self-registration is not possible.

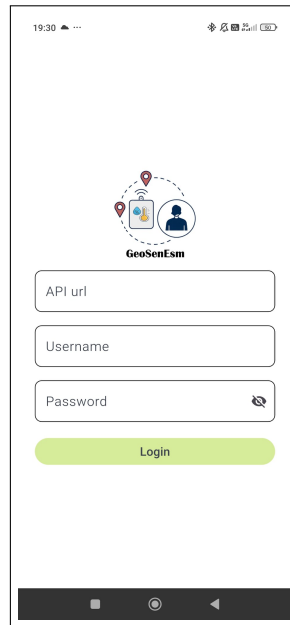


Figure 3.1: *Login screen of the mobile app.*

On the first launch the user is presented with the privacy policy and picks their sensor type (none, Xiaomi or Kestrel). Then the *initial survey* is shown (Figure 3.2); its answers place the respondent in a group that later surveys can target through section-level visibility rules.

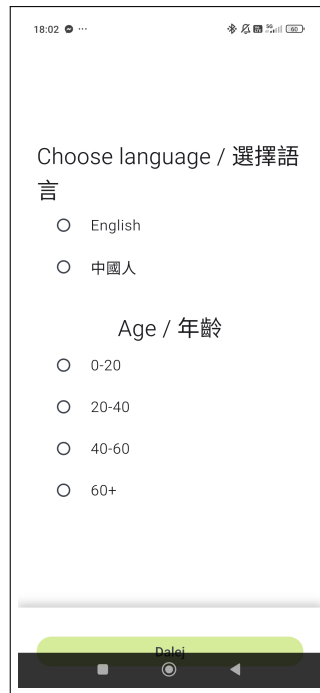


Figure 3.2: Initial-survey example (language and age group).

The app then requests the runtime permissions it needs. All three requests happen on first launch and are shown side by side in Figure 3.3.



Figure 3.3: First-run permission requests.

3.2 Home screen

The home screen (Figure 3.4) is intentionally minimalist. Bright orange buttons show the surveys that are currently open, together with a countdown to the slot's finish time. Navigation to the menu and to the user profile lives in the top-left and top-right corners. The layout respects the device's font-size accessibility setting.

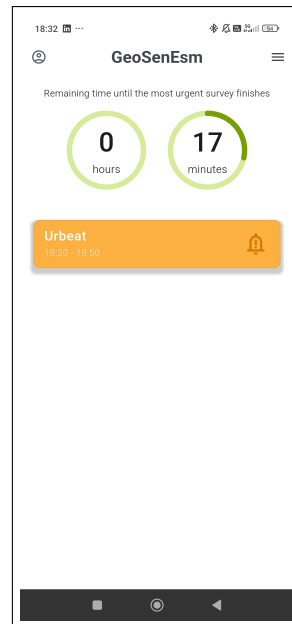
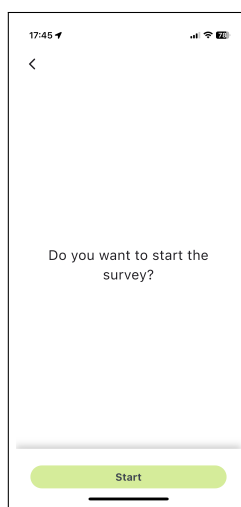


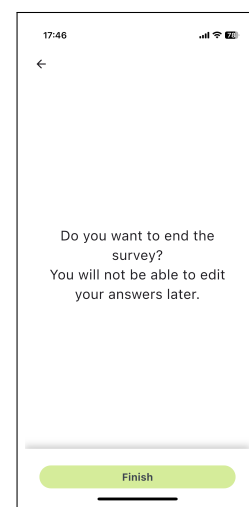
Figure 3.4: Home screen of the mobile app.

3.3 Filling in a survey

1. **Open the app** and tap the orange button whose caption matches the active survey. The countdown next to the button shows how long the slot is still open. Bluetooth should be enabled: the app tries to read the current temperature and humidity from the assigned sensor while the survey is being filled in.
2. **Introduction and outro screens** (Figure 3.5) confirm the start of the survey and the submission of responses. Once submitted, a response cannot be edited.
3. If the respondent chose *no sensor* on the first-run setup, the app additionally asks whether they want to proceed without a sensor reading (Figure 3.6).



(a) Introduction.



(b) Submission confirmation.

Figure 3.5: Confirmation screens shown before and after a survey.

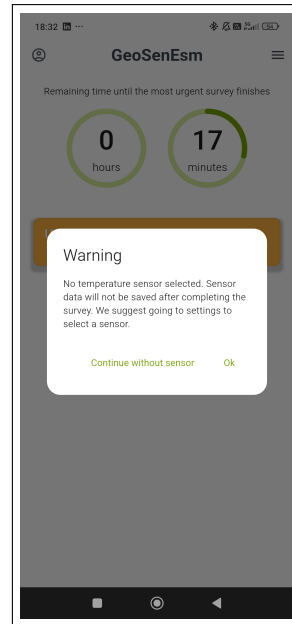
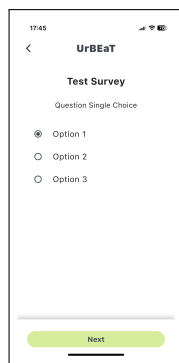


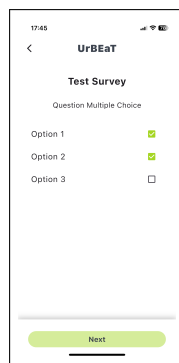
Figure 3.6: “No sensor” warning shown before filling a survey.

3.4 Question types

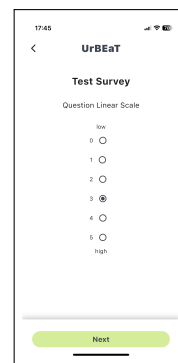
Surveys support seven question types (Figure 3.7).



(a) Single choice.



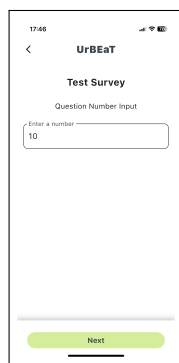
(b) Multiple choice.



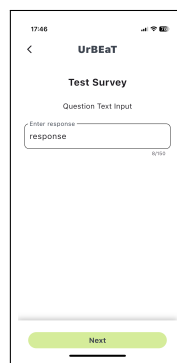
(c) Linear scale.



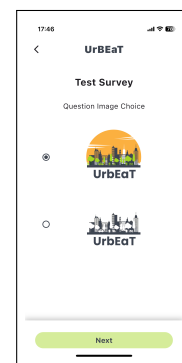
(d) Yes / No.



(e) Number input.



(f) Text input.



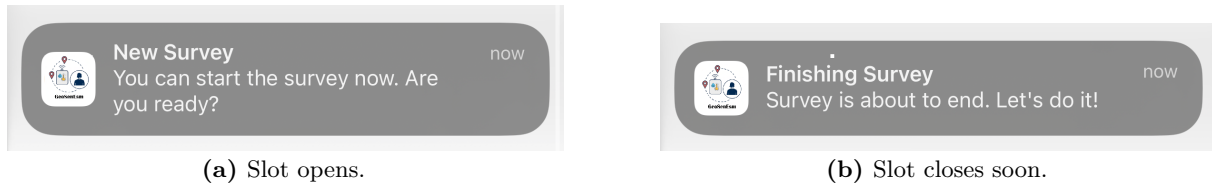
(g) Image choice.

Figure 3.7: The seven question types supported by GeoSenEsm.

On submission the app sends the response to the server together with the matching location and sensor reading.

3.5 Notifications

Respondents are notified when a slot opens and again 15 minutes before it closes (Figure 3.8). The cadence balances respondent engagement against notification fatigue.



(a) Slot opens.

(b) Slot closes soon.

Figure 3.8: Push notifications sent by the mobile app.

3.6 Offline mode

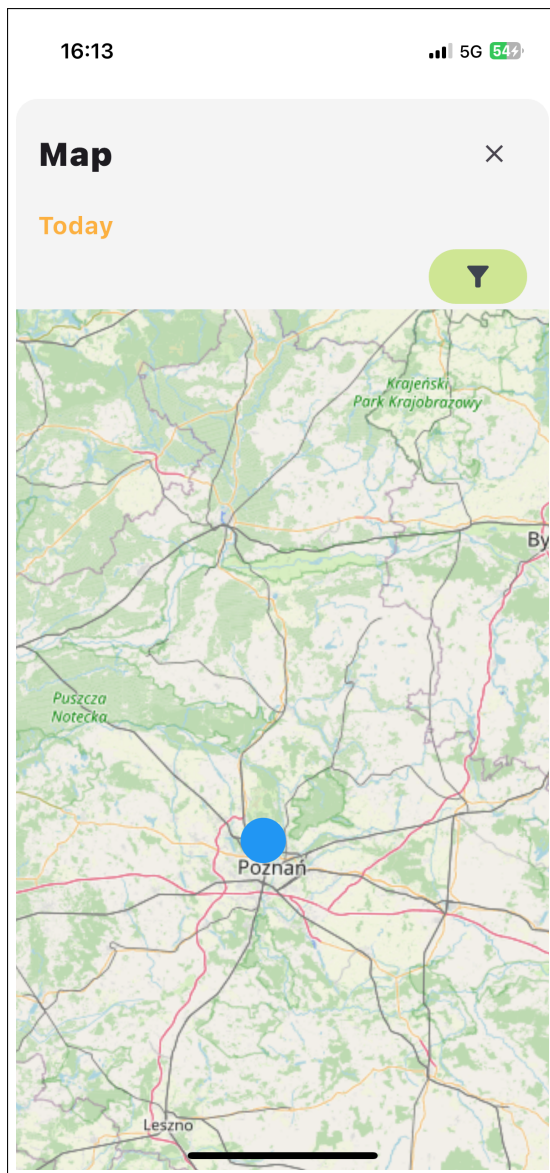
The app works both online and offline. Once a device has synced with the server, all published surveys are cached locally. Respondents can complete them without a network connection; answers, GPS points and sensor readings are pushed to the server as soon as connectivity returns.

3.7 Additional views

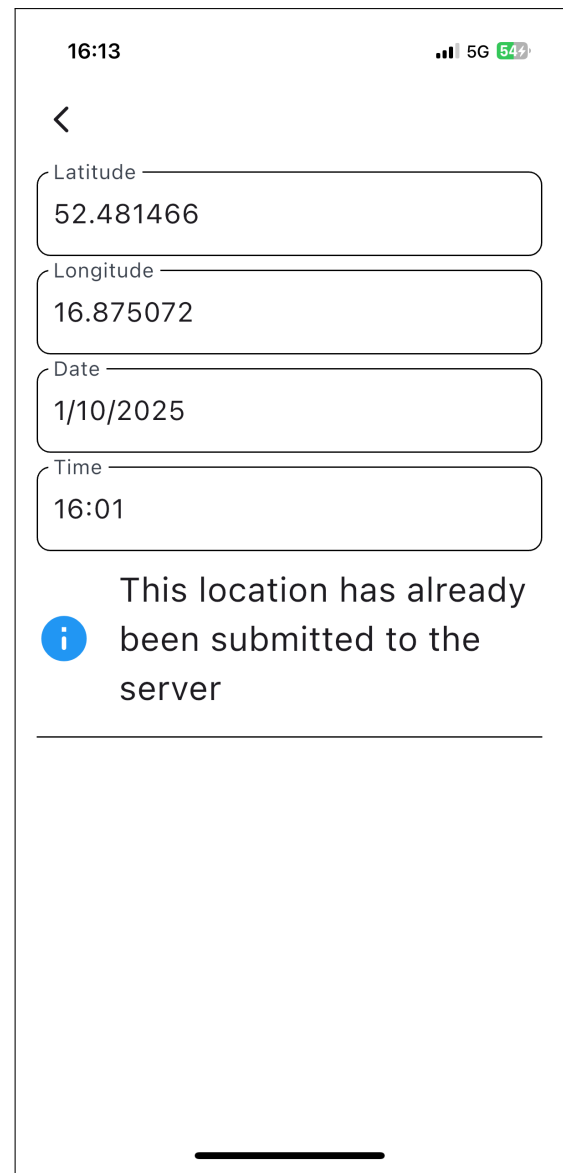
GeoSenEsm exposes what the app has captured to the respondents themselves, building trust and giving them a way to spot problems (for example, a Bluetooth issue that stopped sensor readings for a week).

3.7.1 Map

The map view (Figure 3.9, left) plots every submitted location point. Tapping a marker opens its details (right).



(a) Submitted locations.



(b) Point details.

Figure 3.9: Mobile map module.

3.7.2 Sensor data and history

The sensor module shows live temperature and humidity readings; the history sub-view lists every reading the app has captured (Figure 3.10).

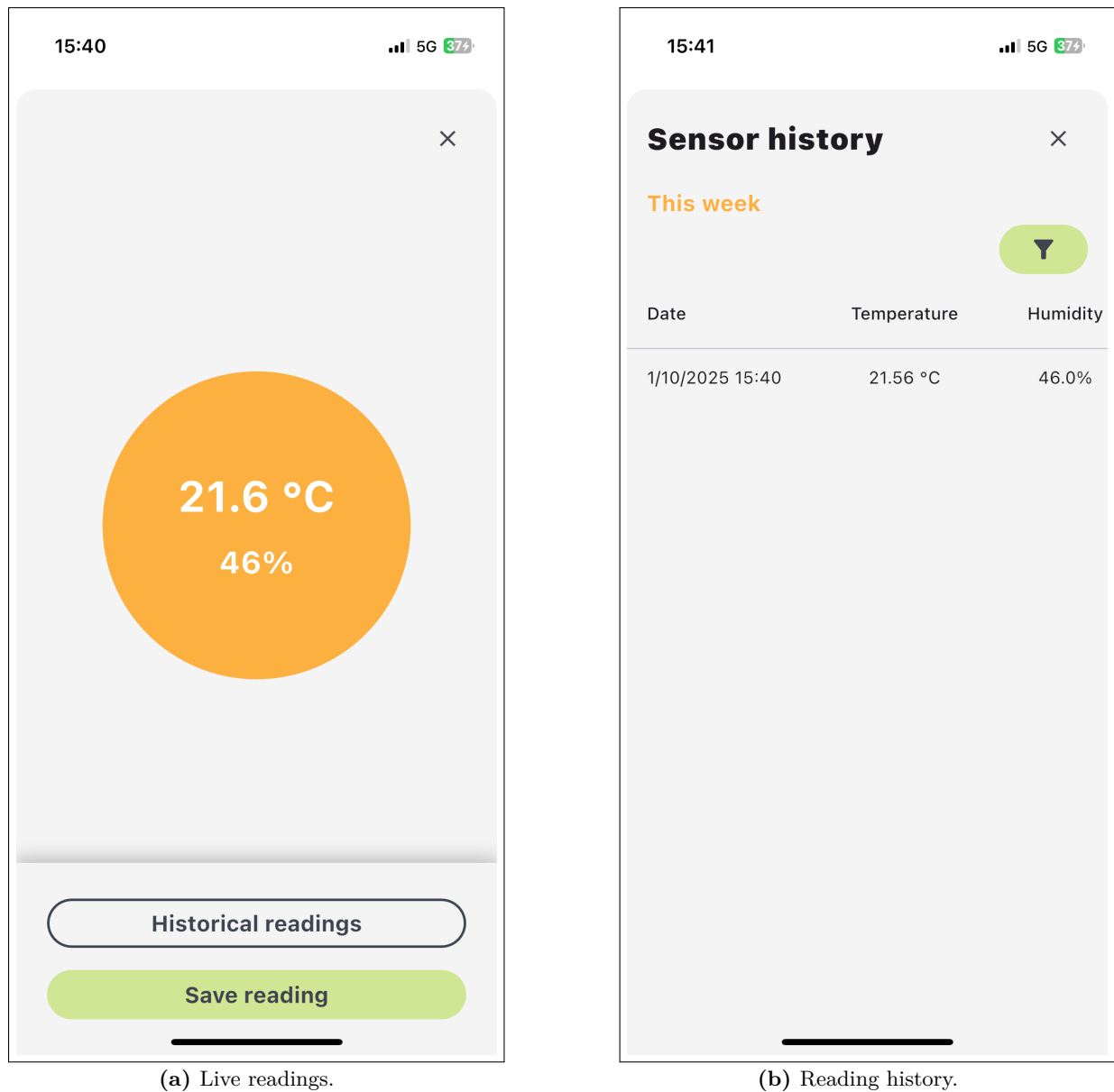


Figure 3.10: *Sensor modules on the mobile app.*

3.7.3 Progress calendar

The calendar (Figure 3.11) colour-codes the respondent's participation history:

- ■ Green — surveys completed on time.
- ■ Red — surveys missed.
- ■ Yellow — surveys still upcoming.

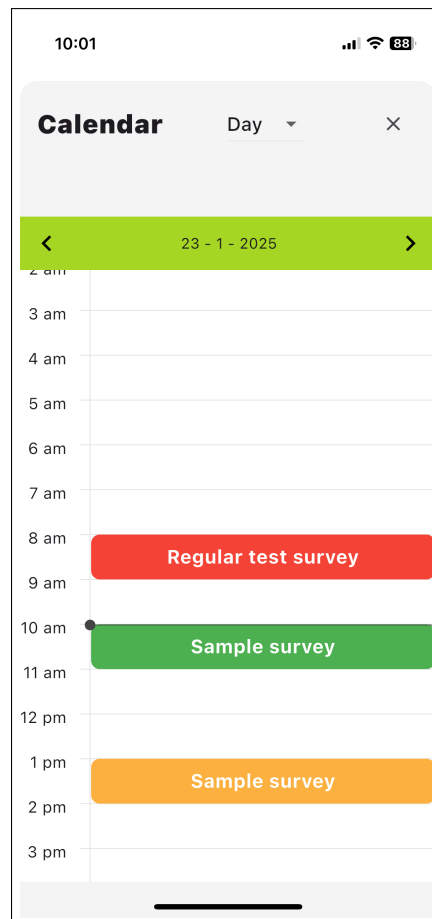


Figure 3.11: Colour-coded participation calendar.

3.8 Design consistency

Surveys deliberately avoid complex gestures such as swiping or drag-and-drop. One question is shown per page and can be answered without scrolling. The colour palette (orange, green, dark grey) and the study logo are used consistently across the mobile app and the admin panel.

4 Admin panel

The admin panel is a browser-based application hosted next to the API. It is where researchers define the study (research area, contact numbers, initial survey, participation slots) and monitor the incoming data (survey results, response documents, statistics, GPS data, sensor readings). Researchers open it in any modern browser at the URL of their deployed instance.

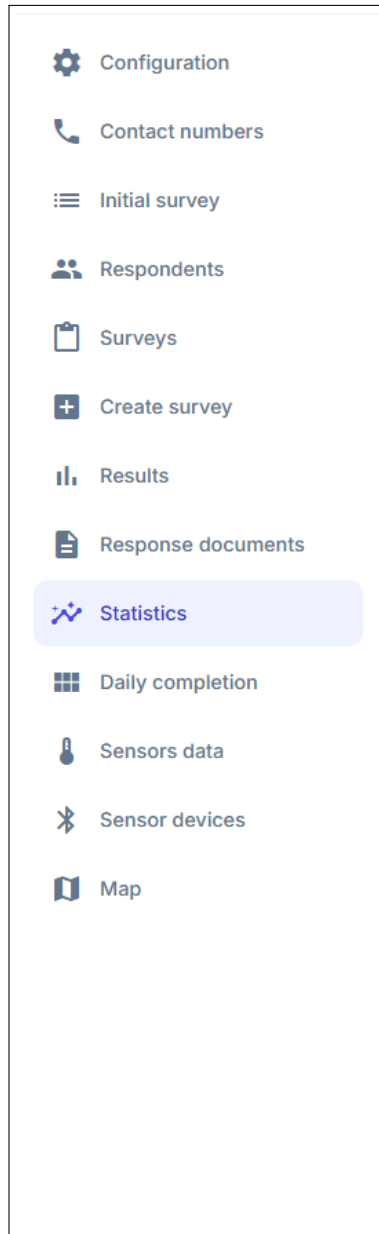


Figure 4.1: *Left-hand navigation of the admin panel.*

The rest of this chapter walks through each section of the navigation.

4.1 Configuration — research area

Researchers can define an optional polygon representing the geographic scope of the study. Every GPS point that arrives from a respondent is compared against this polygon and marked either *inside* or *outside*. The result is used by the map view, by the *Surveys filled outside research area* chart in Statistics (Section 4.9.5) and by the `outsideResearchArea` filter on the GPS-data export.

The polygon is uploaded as a CSV file with latitude / longitude columns. Points must be counter-clockwise and the polygon must be closed (first and last point identical, minimum three unique points).

	A	B
1	longitude	latitude
2	16.811,52.47382065	
3	17.038,52.48762081	
4	17.015,52.35277971	
5	16.785,52.3603283	
6	16.811,52.47382065	

Figure 4.2: Example research-area CSV file.

If the research area changes mid-study, existing GPS rows are re-classified automatically.

4.2 Contact numbers

Phone numbers registered here are shown to the respondent in the mobile app so they can reach the study team (Figure 4.3).

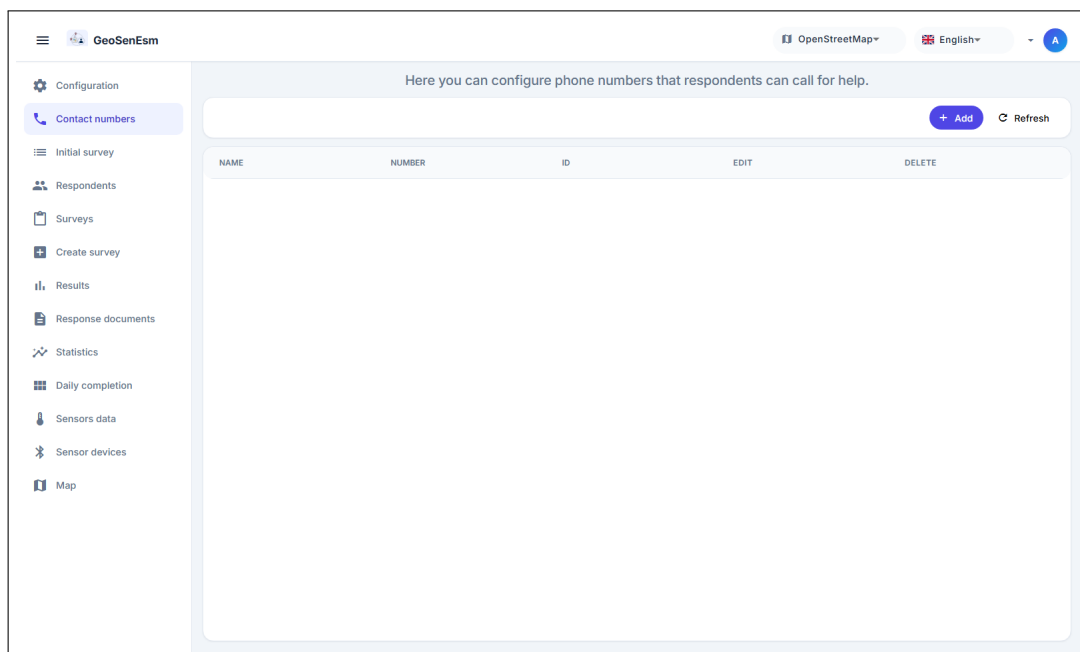


Figure 4.3: Managing study-contact phone numbers.

4.3 Initial survey

The initial survey is shown to the respondent on first login and is mandatory (Figure 4.4). Every initial-survey question must be single-choice — the selected option later becomes a column in the respondent record and can be used to route conditional or group-restricted sections in normal surveys.

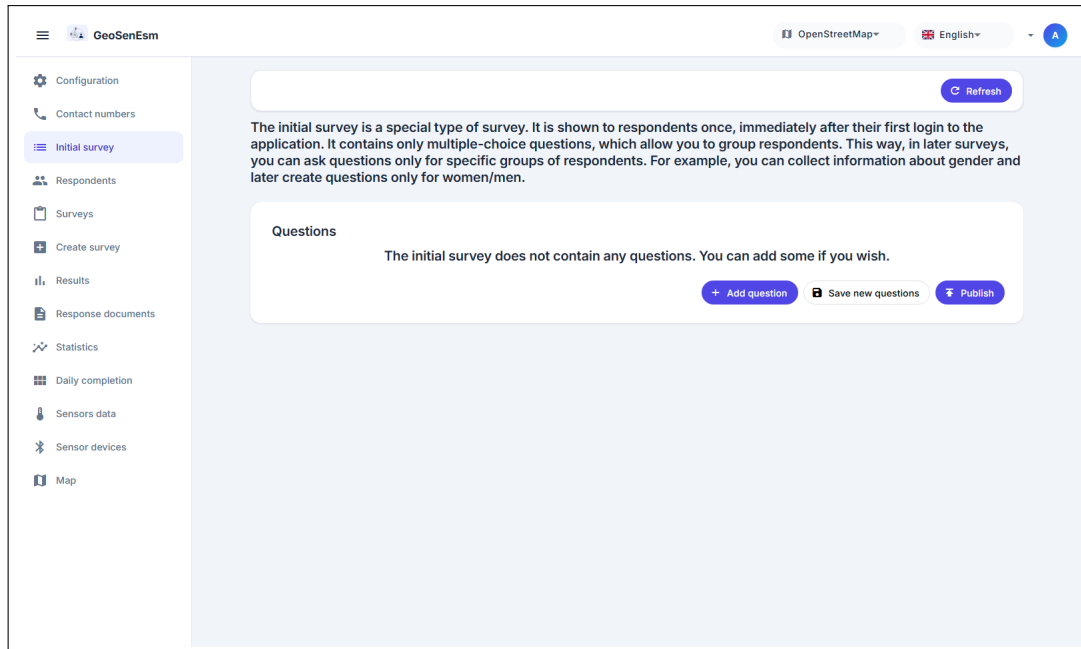


Figure 4.4: *Initial-survey editor.*

4.4 Respondents

The Respondents view lists all accounts, lets the administrator reset a password, and is where new accounts are generated (Figure 4.5).

USERNAME	ID	ACTIONS
00001	b9a05ea4-8748-4a87-add2-be28d795a8c8	[edit] [delete] [icon]
00002	921ca7a5-28b8-44ff-a278-ac2b7c6231e2	[edit] [delete] [icon]
00003	2068451b-0d2c-4a2a-9d10-cfab5c6a0a1	[edit] [delete] [icon]
00004	9e84d095-a8e4-429d-91ae-96555bf75060	[edit] [delete] [icon]
00005	ffb9d939-13e0-46c2-ac02-56dc9e6cf6bb	[edit] [delete] [icon]
00006	e3878578-b2ac-4c6e-a962-23fa62693e3e	[edit] [delete] [icon]
00007	abfdea92-0fd6-4c6b-880d-0d362ceed9d6	[edit] [delete] [icon]
00008	e3ef2220-eb1c-4649-849b-03ece2e7db52	[edit] [delete] [icon]
00009	ab937a5c-119f-46f5-b151-4f193731f23e	[edit] [delete] [icon]
00010	012a7237-9f57-4f3b-b572-e677ecfdc68e	[edit] [delete] [icon]
00011	3d659a0f-c04f-4ee9-a468-c36dabc501dc	[edit] [delete] [icon]

Figure 4.5: Respondents list.

Three advanced filters help find respondents who are not filling the questionnaires actively (Figure 4.6).

Figure 4.6: Advanced respondent filters.

When a batch of accounts is created (Figure 4.7), the generated credentials are copied to the clipboard — they cannot be recovered from the server afterwards. Paste the CSV into a safe place before closing the dialog.

Figure 4.7: Creating respondent accounts. The generated credentials are exported via Copy to clipboard.

4.5 Creating a survey

Survey creation is broken into three steps: editor, sending policy and preview.

4.5.1 Editor

The editor (Figure 4.8) lets the administrator add sections and questions. Each section has one of three visibility rules (Figure 4.9a):

Always visible. Shown to every respondent.

Group visibility. Shown only to respondents whose initial-survey answers place them in a specific group.

Conditional visibility. Shown only if an earlier section's answer matches a rule.

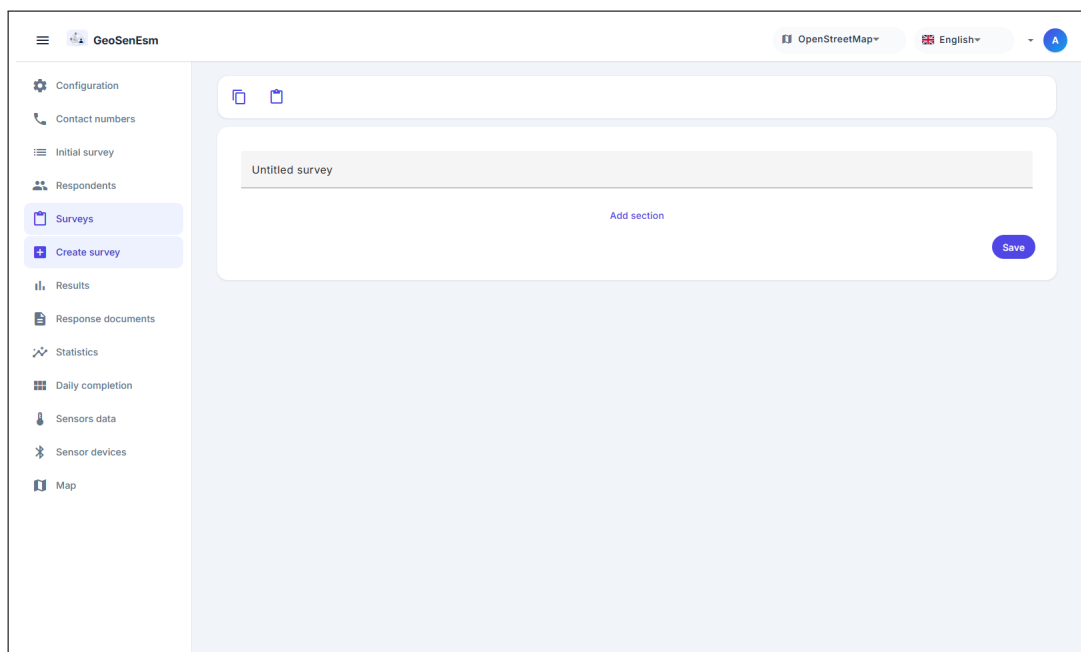
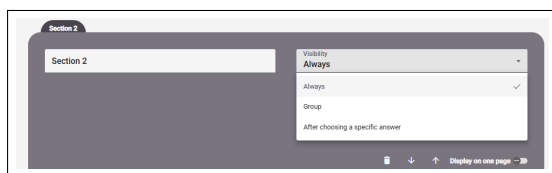
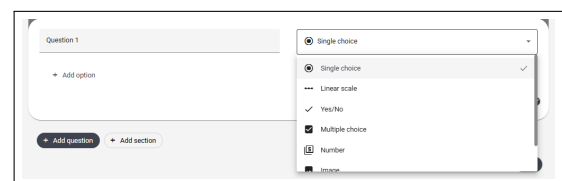


Figure 4.8: Survey editor.



(a) Section visibility rules.

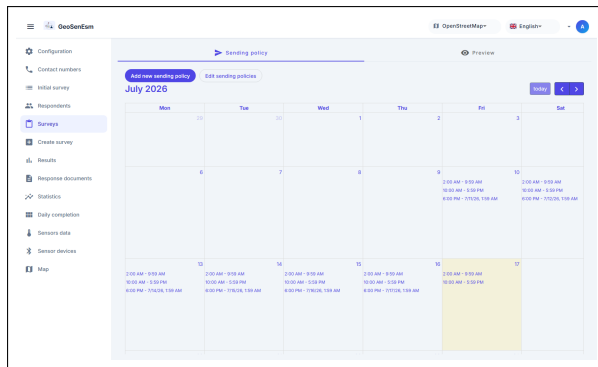


(b) Question types.

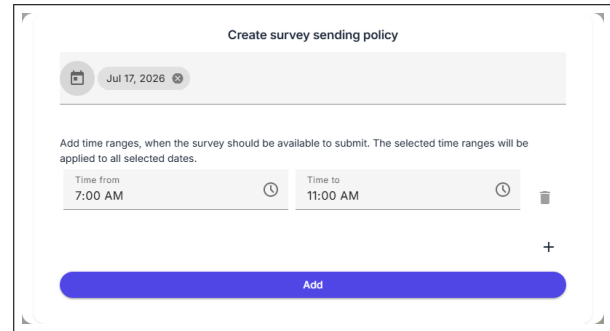
Figure 4.9: Building blocks exposed by the editor.

4.5.2 Sending policy

Once the structure is saved, the administrator defines the sending policy — when the survey will be open for participants. The schedule is drawn on a calendar view (Figure 4.10, left); the per-slot form is shown on the right.



(a) Calendar overview.

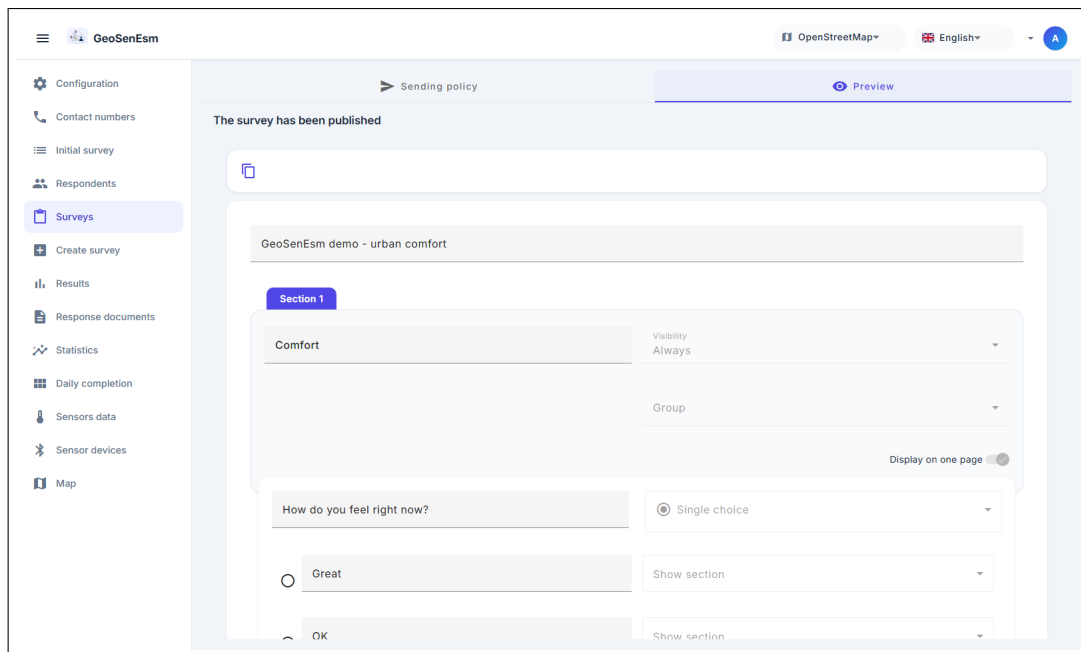


(b) Per-slot form.

Figure 4.10: *Sending-policy editor.*

4.5.3 Preview and publish

The preview tab (Figure 4.11) is a final opportunity to spot mistakes. Once the survey is published, its structure becomes immutable — respondents can start filling it in, and further edits are no longer possible. The preview also offers a *Copy* action which is handy when reusing a complicated structure as the basis for a new survey.

Figure 4.11: *Survey preview.*

4.6 Surveys list

The Surveys tab lists every survey together with its current sending policy. Schedules can be edited from here at any time.

4.7 Results

The Results tab (Figure 4.12) shows a tabular view of all submitted answers. Rows can be filtered by survey and by respondent and exported to CSV.

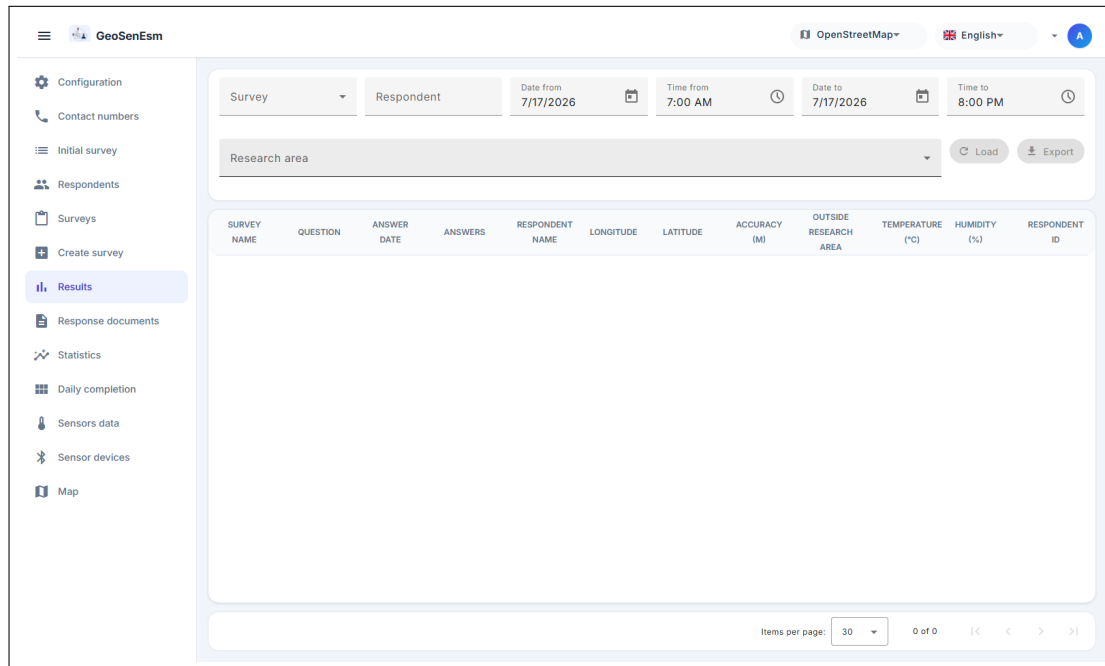


Figure 4.12: Results tab.

4.8 Response documents

The Response documents tab is backed by MongoDB. It contains one JSON document per submission, with the exact payload the mobile app sent to the server (answers, timestamps, embedded sensor reading, matching survey metadata). Two workflows are supported:

- **Download** one specific document from the grid by clicking the download icon in the row. The result is a self-contained JSON file that can be inspected offline.
- **Export** the full filtered set as a ZIP archive.

By default the date range is set to *today* → *tomorrow* (local time), which matches the most common use case: reviewing the responses that have just come in (Figure 4.13). The *Clear filters* button resets the range back to that default. The paginator and page-size selector stay visible even when the results table is very wide, so switching pages does not require horizontal scrolling.

PARTICIPATION DATE	SURVEY	RESPONDENT	ANSWERS	ACTIONS
7/17/26, 4:39 PM	GeoSenEsm demo - urban comfort	00028	4	Download
7/17/26, 3:56 PM	GeoSenEsm demo - urban comfort	00024	4	Download
7/17/26, 3:15 PM	GeoSenEsm demo - urban comfort	00033	4	Download
7/17/26, 2:53 PM	GeoSenEsm demo - urban comfort	00020	4	Download
7/17/26, 12:38 PM	GeoSenEsm demo - urban comfort	00032	4	Download
7/17/26, 12:32 PM	GeoSenEsm demo - urban comfort	00017	4	Download
7/17/26, 11:46 AM	GeoSenEsm demo - urban comfort	00016	4	Download
7/17/26, 11:36 AM	Demo 1136	00015	3	Download
7/17/26, 11:36 AM	Demo 1136	00014	3	Download
7/17/26, 11:36 AM	Demo 1136	00013	3	Download
7/17/26, 11:06 AM	GeoSenEsm demo - urban comfort	00036	4	Download
7/17/26, 10:56 AM	GeoSenEsm demo - urban comfort	00025	4	Download
7/17/26, 10:26 AM	GeoSenEsm demo - urban	00040	4	Download

Figure 4.13: Response documents tab (Mongo-backed).

4.9 Statistics

The Statistics section aggregates the operational data across the whole study and is the fastest way to get a feel for how the study is going. It is split into three view-toggles at the top of the page (*Global*, *Daily*, *Per participant*) and a dedicated *Daily completion* entry in the side menu. Every view shares a common KPI strip and a set of time-series charts underneath.

Whenever data is not yet available from the backend, a spinner is shown in place of the KPIs and charts. Switching views clears the stale values so the spinner is visible during every refresh.

4.9.1 Global view

Aggregates the whole study window into per-day time series (filled surveys, GPS points, sensor readings, participations outside the research area). It is the right place to look for weekly trends or unusual dips in data collection.

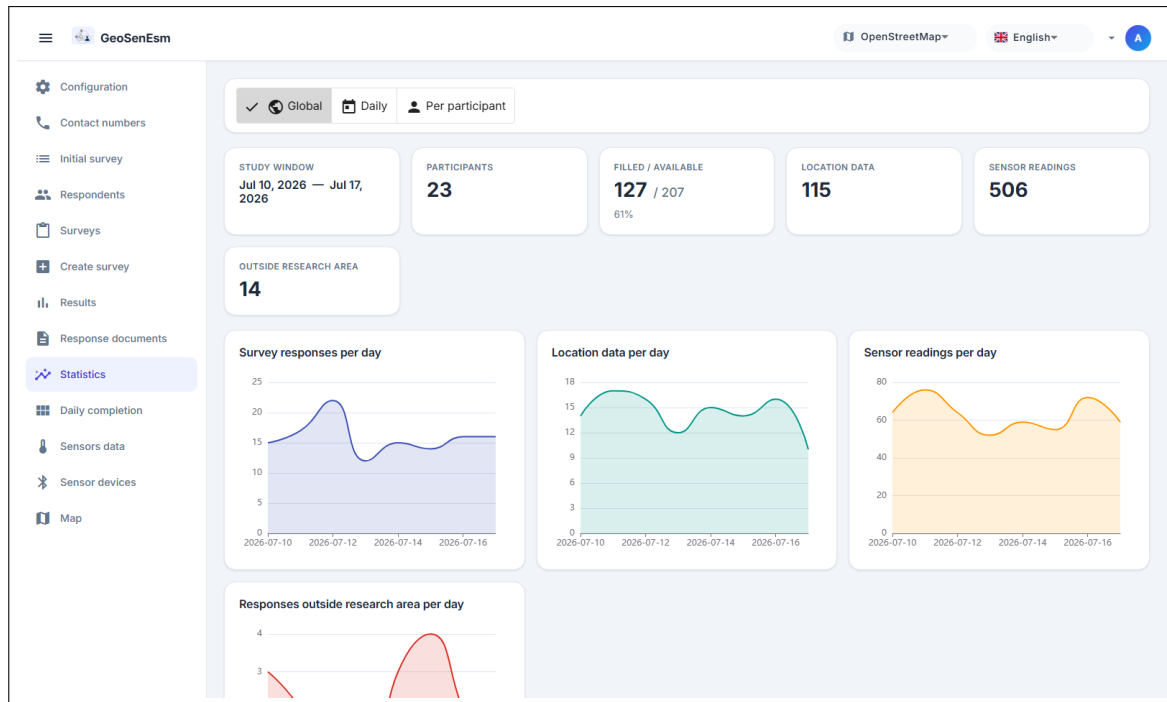


Figure 4.14: Statistics — Global view.

4.9.2 Per participant view

Focuses on a single respondent. The participant is picked from a **type-to-filter** input — type a substring of a username and the dropdown narrows to the matching accounts. Once a participant is chosen, the KPI strip and per-day time series update accordingly (Figure 4.15).

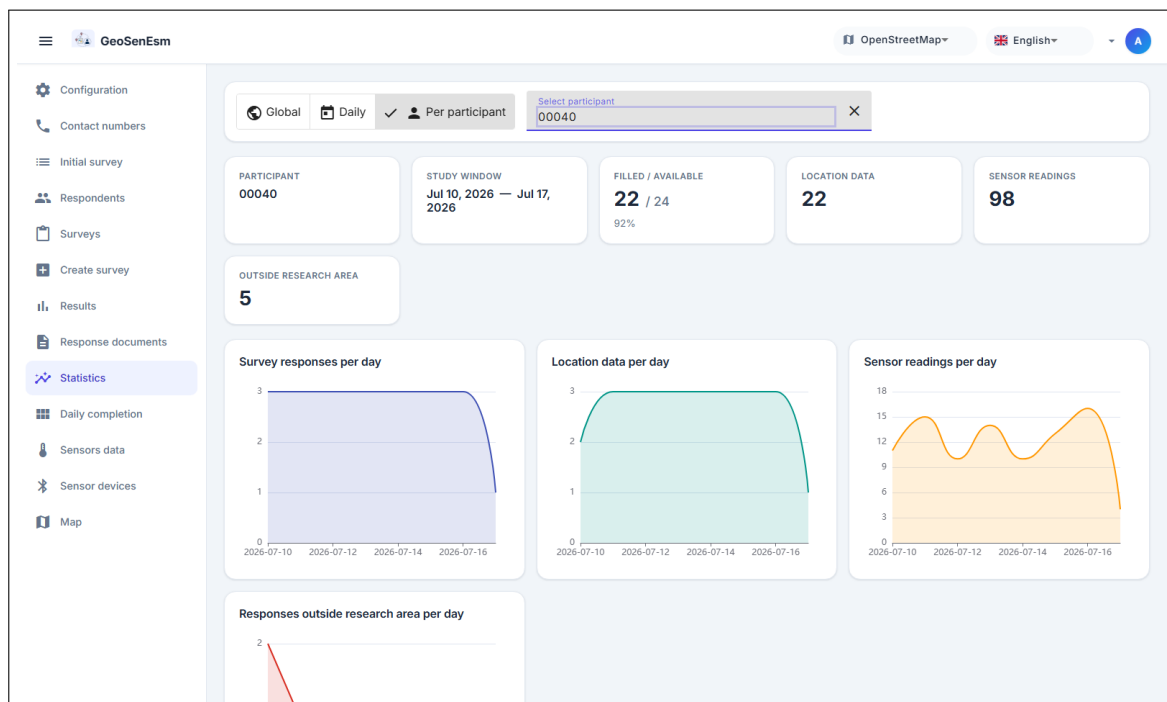


Figure 4.15: Statistics — Per participant view with username autocomplete.

4.9.3 Daily view

Zooms into a single UTC calendar day. Instead of per-day series, the charts show per-hour buckets, which is useful when a slot schedule places several surveys inside the same day (Figure 4.16). The daily view also exposes an extra KPI, *Filled / Available active*, which counts only respondents that have submitted at least one survey inside the trailing 3-day window ending at the selected day. Hovering the label opens a tooltip that explains the definition.

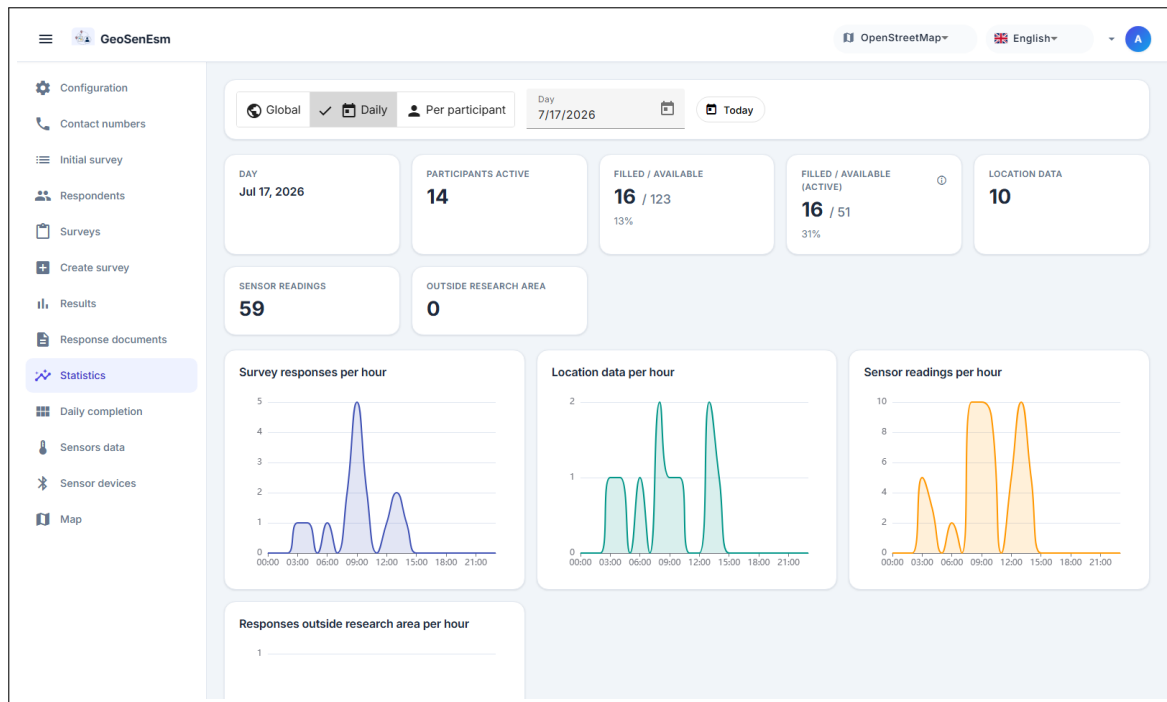


Figure 4.16: *Statistics — Daily view with hourly buckets and the “Active” KPI.*

4.9.4 Daily completion

Daily completion (Figure 4.17) is the compliance view. For a chosen day it draws one small card per respondent with a coloured stripe for every scheduled slot. The colour encodes what has been collected against that slot:

Daily-completion colour legend

- Submission collected with GPS *and* sensor data.
- Submission collected with GPS only.
- Submission collected with sensor only.
- Submission collected but neither GPS nor sensor data was linked.
- Slot has finished and no submission arrived.
- Slot has not started yet.

Each participant card can be resized: the toolbar exposes two sliders (width 110–220 px, height 90–160 px, step 5 px) with the selected size persisted in `localStorage` so the layout is stable across sessions.

The toolbar additionally offers a *Only active* filter with an adjustable window (default 7 days), see Figure 4.18. When checked, the grid hides respondents whose last submission is older than N days, measured *relative to the day chosen in the calendar* — i.e., the window is $[\text{end-of-day} - N \text{ days}, \text{end-of-day}]$, so inspecting a past day never counts submissions that arrived after it.

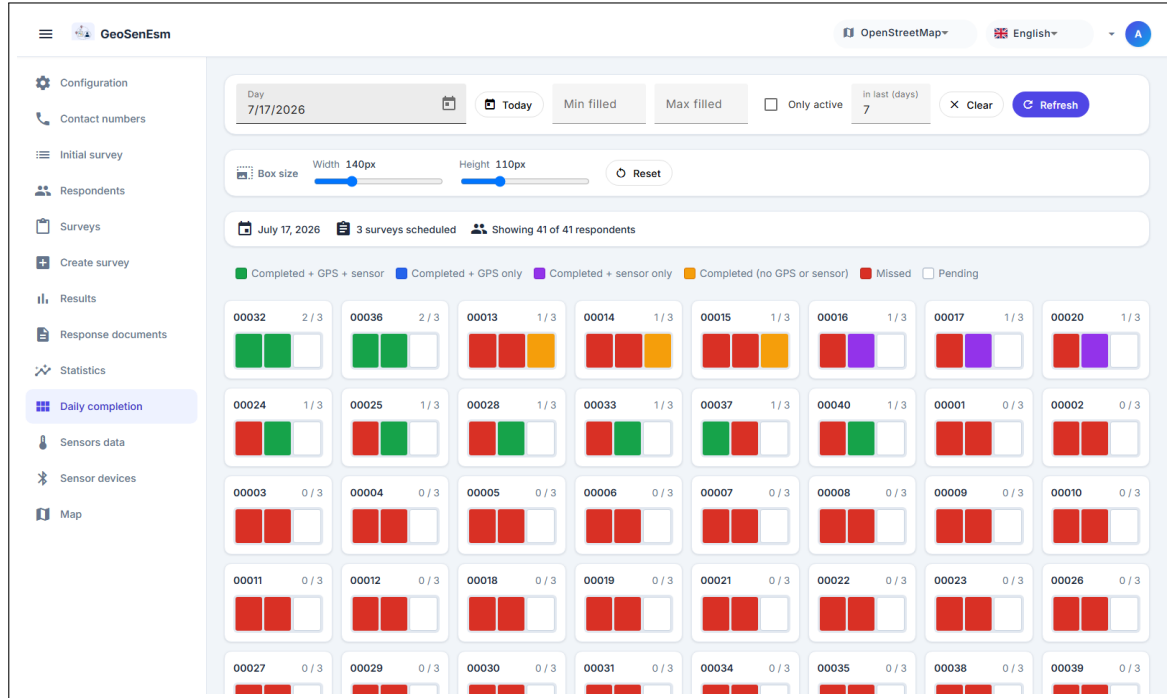


Figure 4.17: *Statistics — Daily completion.*

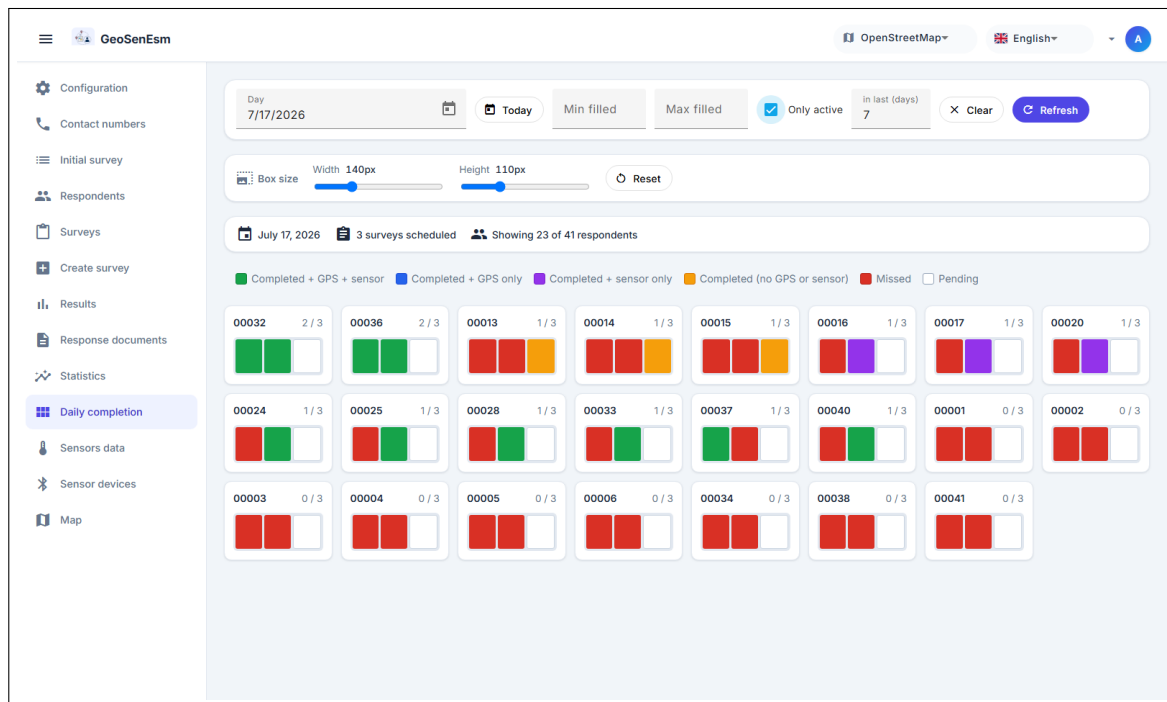


Figure 4.18: *Daily completion with the “Only active in last N days” filter engaged — inactive respondents disappear from the grid.*

4.9.5 Surveys outside research area

When a research-area polygon is defined, every GPS point that arrives is compared against it. Statistics includes a dedicated KPI card and time series that count submissions whose linked GPS row falls outside the polygon (Figure 4.19). Present on the Global, Daily and Per participant views, this number is a quick sanity check that respondents actually stayed within the study area.

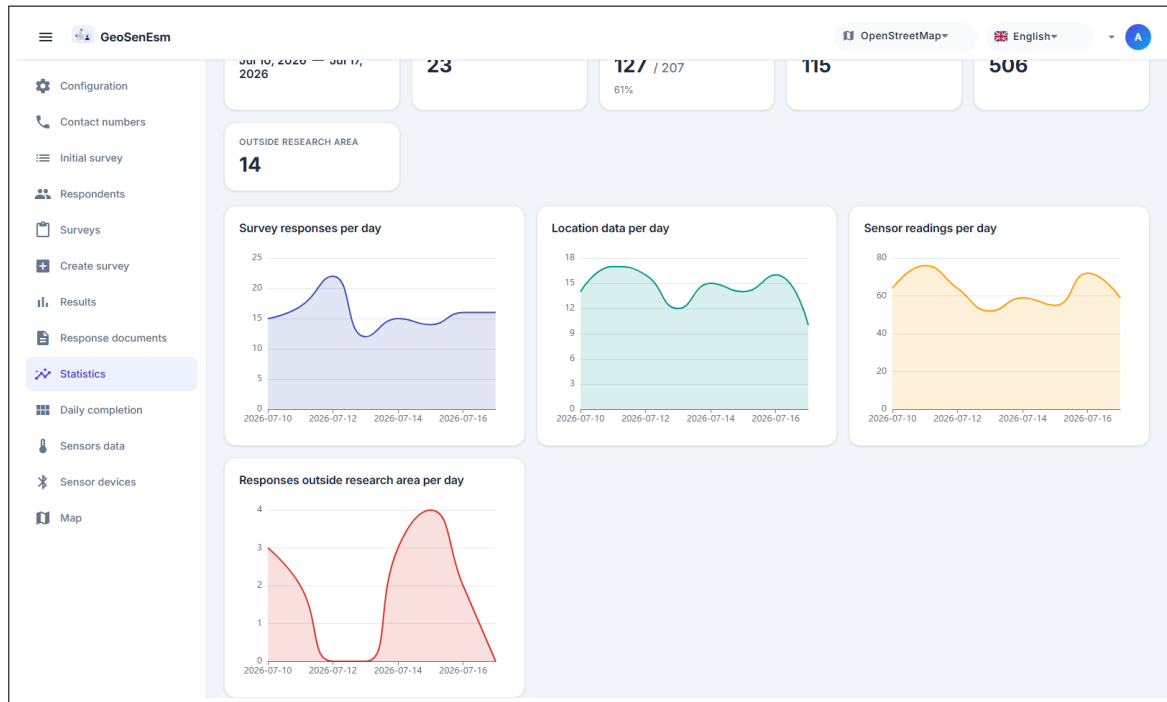


Figure 4.19: “Surveys filled outside research area” KPI and chart.

4.10 Sensor data

Every sensor reading uploaded by respondents can be inspected here (Figure 4.20). Filters mirror the ones available for GPS data (date range, respondent).

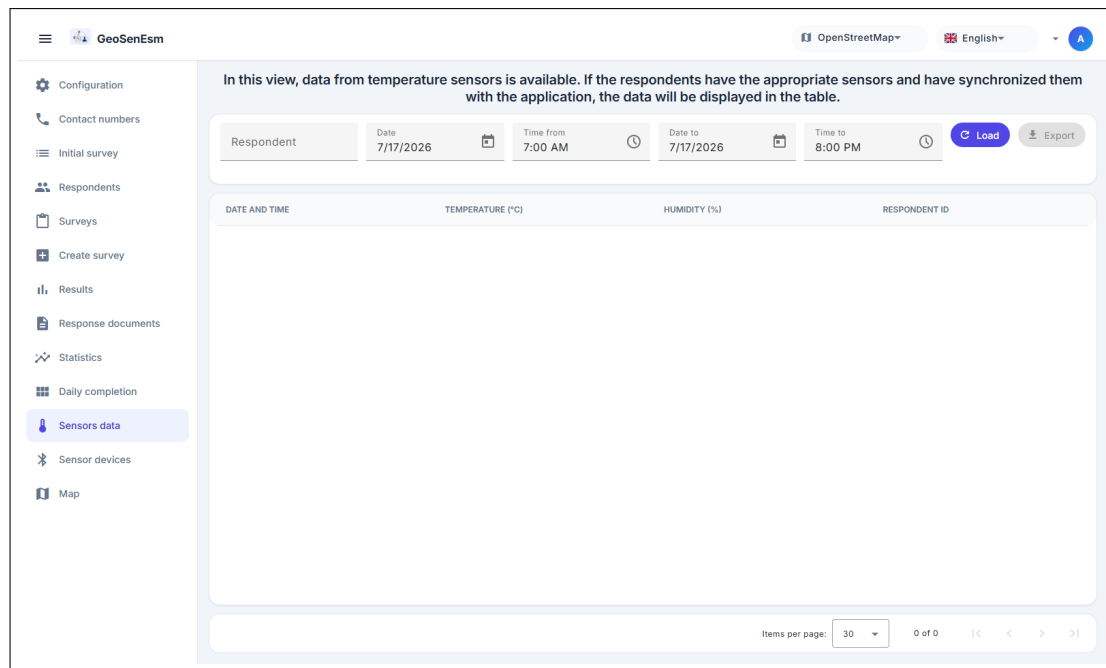


Figure 4.20: Sensor data tab.

4.11 Sensor devices

The Sensor devices tab (Figure 4.21) is where a MAC address can be tied to a Xiaomi sensor number. This lets the mobile app connect only to *that* respondent's device, which is important because every Xiaomi LYWSD03MMC broadcasts the same Bluetooth name. Kestrel devices are uniquely identifiable out of the box, so the whitelist is not required for them.

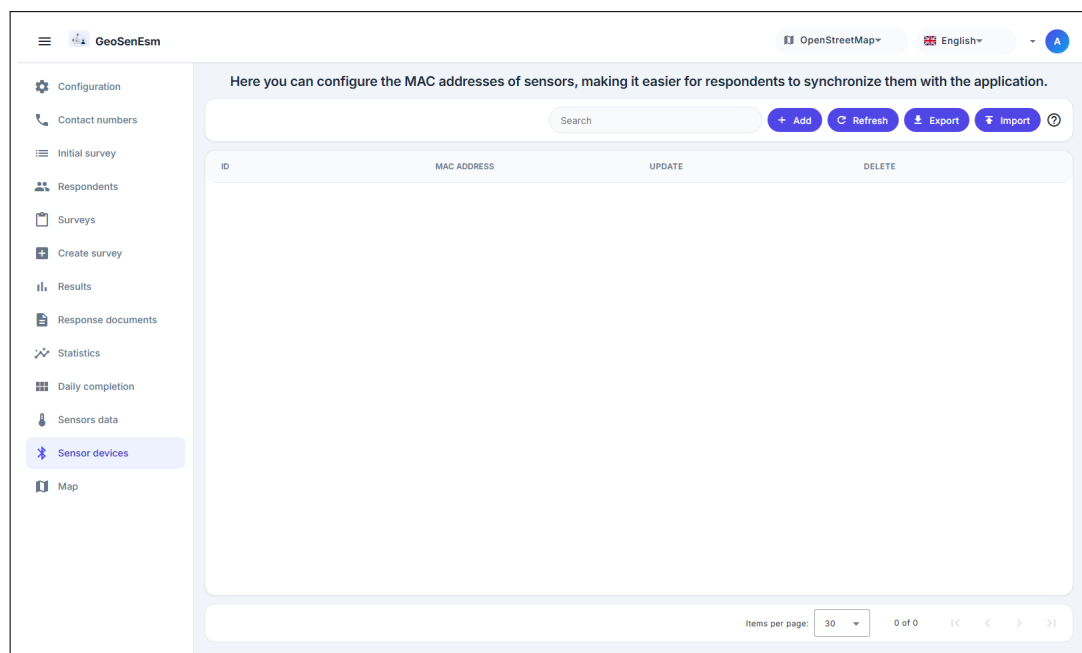


Figure 4.21: Xiaomi sensor MAC whitelist.

4.12 Map

The Map tab (Figure 4.22) plots every GPS point in the system. Points are colour-coded by whether they lie inside or outside the research-area polygon.

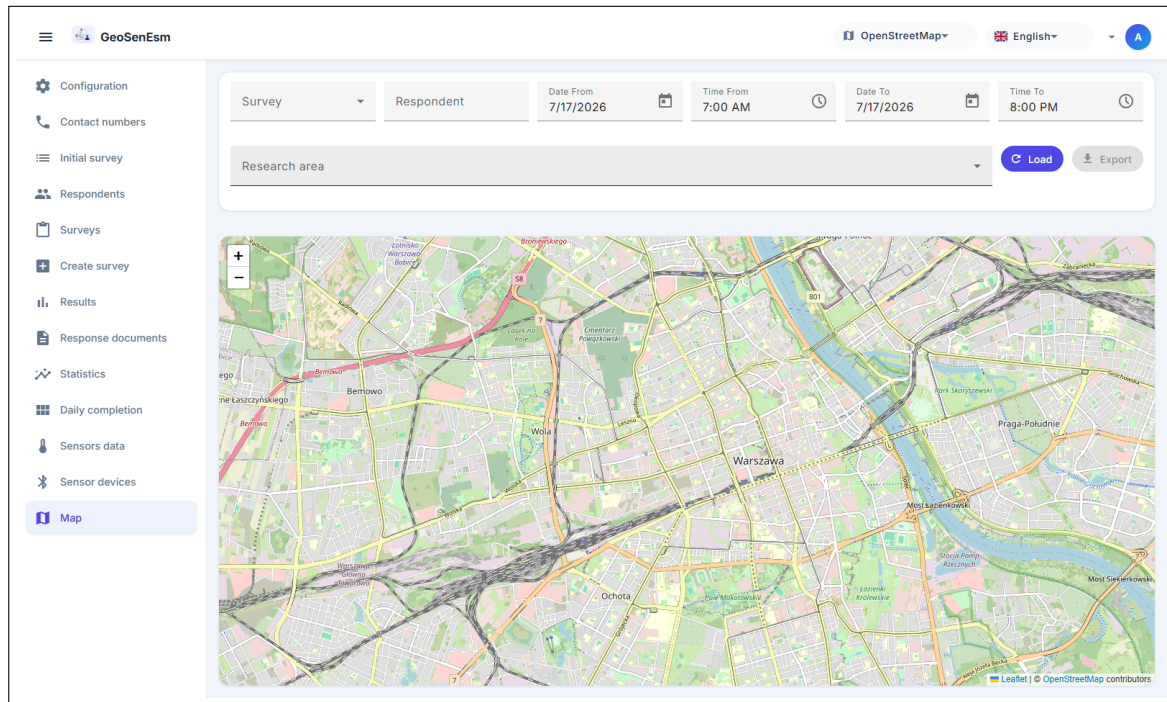


Figure 4.22: Map tab — GPS points shaded by inside vs. outside the research area.

5 Closing notes

The design of GeoSenEsm follows a few consistent principles:

- *Prioritise the respondent.* Keep the mobile UI simple, the permissions transparent and the sensitive data controllable.
- *Meet the researcher where they are.* Every artefact (results, response documents, GPS, sensor readings) is exportable to CSV, JSON or ZIP so it can flow straight into existing analysis pipelines.
- *Make problems easy to spot.* The Statistics dashboard, the Daily completion grid, the *Only active* filter and the outside-research-area KPI are all there so an issue with a single respondent or a single day of the study is visible at a glance.

The Experience Sampling Method integrates naturally with a mobile application: participants carry their phone with them at all times, and the app can capture context (location, temperature, humidity) alongside their subjective response. Cross-referencing the two lets researchers build a richer picture of how urban environments shape thermal comfort and everyday behaviour.